Maxima and Minima: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Maxima and minima at a point

Formulas

Maxima at a: $f(a) > f(a \pm h)$; minima at a: $f(a) < f(a \pm h)$ (small $h > 0$)

KEY POINTS

- Maxima / minima are points of the domain
- Continuity / differentiability not needed to define them
- End point: compare only with the side inside the domain

2. Words and important facts

KEY POINTS

- Maxima = maximum = local / relative maxima; minima likewise
- Greatest value = global / absolute maximum; least value = global / absolute minimum
- Extrema = extremum = extremal points = turning points
- Value at a maxima may not be the greatest value; a maximum value can be less than a minimum value
- Continuous non-constant f: maxima and minima alternate

3. Greatest and least value on $[a, b]$

STEPS

1. Find critical points in (a, b): $f' = 0$ or f' does not exist
2. Evaluate f there and at a, b
3. Largest = greatest value, smallest = least value

Watch out

With | | or [ ] include the points where the expression inside is 0 / an integer

4. First derivative test

Formulas

f': $+ \to -$ maxima; $- \to +$ minima; no change: neither

STEPS

1. Domain
2. Critical points ($f' = 0$ or not defined, in the domain)
3. Sign scheme of f'; even powers do not change sign
4. Read maxima / minima

KEY POINTS

- Works also where f' does not exist (corners, $|x|$)
- $f = \int_a^x \phi(t)\,dt \Rightarrow f' = \phi(x)$; $\ln g = f \Rightarrow g$ and f have the same extrema

5. Second and nth derivative tests

Formulas

$f'(a) = 0$: $f''(a) > 0$ minima, $f''(a) < 0$ maxima, $f''(a) = 0$ test fails
First non-zero $f^{(n)}(c)$: n even: < 0 max, > 0 min; n odd: no extremum

Watch out

Second derivative test needs $f'(a) = 0$ and $f''(a)$ to exist

6. Concavity

Formulas

$f'' > 0$: concave up, f' increasing, tangent below the curve, chord above
$f'' < 0$: concave down, f' decreasing, tangent above the curve, chord below

KEY POINTS

- Concave up: $f\left(\frac{x_1+x_2}{2}\right) < \frac{f(x_1)+f(x_2)}{2}$; concave down: reverse
- Naming differs: the sheet calls $f'' > 0$ 'concave' and $f'' < 0$ 'convex'

7. Point of inflection

KEY POINTS

- Tangent exists and concavity changes
- $f'' = 0$ or does not exist, and f'' changes sign
- Inflection points of f = extremum points of f'
- From the graph of f': rising / falling parts give concave up / down

Watch out

$f''(a) = 0$ alone is not inflection (x^4 at 0); horizontal tangent + inflection: $f' = f'' = 0$

8. Geometrical and practical problems

STEPS

1. Figure; constants vs variables
2. Quantity in one variable via the given relation
3. $f' = 0$, sign / f''; check end values of the allowed interval

Formulas

Cone: $V = \frac{1}{3}\pi r^2 h$, curved surface $\pi r l$; cylinder: $V = \pi r^2 h$, curved $2\pi r h$
Sphere: $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$; sector: area $\frac{1}{2}r^2\theta$, arc $r\theta$

KEY POINTS

- Use an angle as the variable for figures in circles / ellipses ($x = a\cos\theta$, $y = b\sin\theta$)
- Maximise the square of a root expression (area^2) to avoid roots

9. Analysis of a cubic

Formulas

$f = ax^3 + bx^2 + cx + d$, $f' = 3ax^2 + 2bx + c$, $f'' = 0 \Rightarrow x = -\frac{b}{3a}$
Roots $\alpha < \beta$ of $f' = 0$: $f(\alpha)f(\beta) < 0$ three, $= 0$ two (one repeated), > 0 one real root

KEY POINTS

- $D(f') \leq 0$: monotonic, exactly one real root, no extremum
- $a > 0$: maxima at α (smaller), minima at β (larger); $a < 0$: reverse
- Sign of maxima / minima point = location of roots of the quadratic $f' = 0$
- $x^3 + px + q$: three distinct roots iff $4p^3 + 27q^2 < 0$

10. Polynomial from given extrema

STEPS

1. $f'(x) = k(x-\alpha)(x-\beta)\cdots$ or general polynomial
2. Finite $\lim_{x\to 0}\frac{f(x)}{x^m}$: lower coefficients are 0
3. 'f' has extremum at c' $\Rightarrow f''(c) = 0$; use given values

11. Common mistakes checklist

STEPS

1. Forgetting points where f' does not exist
2. Calling every $f' = 0$ point an extremum (check the sign change)
3. Not checking end points for greatest / least value
4. Taking a point outside the domain as maxima / minima
5. $f'' = 0$ taken as inflection without sign change
6. Cubic with $a < 0$: maxima and minima roles swap